

E1-02 Prior Work

source: research/prior_work.csv — v1.4.0 (2026-09-22)

(12 fields; rendered as field dictionaries to preserve all columns)

Row 1

source_id	PW01
title	A Theory for Record Linkage
authors	Fellegi I.P.; Sunter A.B.
year	1969
url_doi	https://doi.org/10.1080/01621459.1969.10501049
reading_scope	bib-and-abstract-verified
problem_addressed	Optimal decision rule for linking records across files
data_method	Decision-theoretic model of comparison vectors with link/non-link/possible-link dispositions
limitations	Assumes comparison-pair errors follow known m/u probabilities; needs calibration
overlap_with_candidates	Theoretical basis for constrained linkage decision rule and needs-review class
proposed_difference	Explicit uncertainty-preserving tier (needs_review) with downstream conflict checks instead of forced dispositions
verifiable_location	JASA 64(328):1183-1210

Row 2

source_id	PW02
title	Data Matching: Concepts and Techniques for Record Linkage Entity Resolution and Duplicate Detection
authors	Christen P.
year	2012
url_doi	https://doi.org/10.1007/978-3-642-31164-2
reading_scope	bibliography-verified
problem_addressed	Broad survey of modern record linkage blocking similarity and evaluation
data_method	Survey of matching techniques blocking methods and evaluation measures
limitations	Generic survey; no billing-domain event semantics
overlap_with_candidates	Background for candidate blocking and precision/recall evaluation
proposed_difference	Domain-specific event dictionary and lineage semantics on top of generic matching
verifiable_location	Springer 2012

Row 3

source_id	PW03
title	Process Mining: Data Science in Action

authors	van der Aalst W.
year	2016
url_doi	https://doi.org/10.1007/978-3-662-49851-4
reading_scope	bibliography-verified
problem_addressed	Event log analysis conformance and discovery
data_method	Process mining over event logs with timestamps
limitations	Assumes process logs; billing events are messier multi-source
overlap_with_candidates	Framework for event normalization and reconstruction audit
proposed_difference	Explicit handling of late-arriving records and as-of truncation for point-in-time correctness
verifiable_location	Springer 2016 (2nd ed.)

Row 4

source_id	PW04
title	A Proportional Hazards Model for the Subdistribution of a Competing Risk
authors	Fine J.P.; Gray R.J.
year	1999
url_doi	https://doi.org/10.1080/01621459.1999.10474144
reading_scope	bib-and-abstract-verified
problem_addressed	Regression on cumulative incidence under competing risks
data_method	Semiparametric subdistribution hazard model
limitations	Continuous-time hazard formulation
overlap_with_candidates	Statistical grounding for competing payment/closure events
proposed_difference	Discrete-time interval-based conditional probabilities suitable for irregular observation
verifiable_location	JASA 94(446):496-509

Row 5

source_id	PW05
title	Nonparametric Estimation from Incomplete Observations
authors	Kaplan E.L.; Meier P.
year	1958
url_doi	https://doi.org/10.1080/01621459.1958.10501452
reading_scope	bibliography-verified
problem_addressed	Survival estimation with right censoring
data_method	Product-limit estimator
limitations	Classical; no covariates
overlap_with_candidates	Censoring conventions adopted in v1
proposed_difference	H-dependent interval decomposition with explicit open-state probability
verifiable_location	JASA 53(282):457-481

Row 6

source_id	PW06
title	Deep Claim: Payer Response Prediction from Claims Data with Deep Learning
authors	Kim B.-H.; Sridharan S.; Atwal A.; Ganapathi V.
year	2020
url_doi	https://arxiv.org/abs/2007.06229
reading_scope	bib-and-abstract-verified
problem_addressed	Denial and response-date prediction from claims
data_method	Deep multi-task architecture over 2.9M claims from two US health systems
limitations	Pre-submission features; no billing-event reconstruction; deep model not needed for this study
overlap_with_candidates	Closest applied work on payer response prediction
proposed_difference	Adds explicit reconstruction-quality features and uncertainty-preserving linkage as inputs; simpler transparent baseline comparison
verifiable_location	arXiv:2007.06229 (ICML 2020 HSYS Workshop)

Row 7

source_id	PW07
title	Predicting Invoice Payment Outcomes for Optimized Collections
authors	Zeng S.; Melville P.; Lang C.A.; et al.
year	2007
url_doi	https://prem-melville.com/publications/equitant-kdd07.pdf
reading_scope	bib-and-abstract-verified
problem_addressed	Classify invoice payment outcome into delay buckets for collection prioritization
data_method	Supervised learning over invoice-level features
limitations	Generic B2B invoices; no claim adjudication semantics
overlap_with_candidates	Baseline inspiration for delay-bucket prediction and aging features
proposed_difference	Event-level observation-window aware labeling instead of static outcome buckets
verifiable_location	KDD 2007 Workshop on Data Mining for Business Applications

Row 8

source_id	PW08
title	A machine-learning approach towards solving the invoice payment prediction problem
authors	Schoonbee S.
year	2022
url_doi	https://scielo.org.za/scielo.php?pid=S2224-78902022000400010&script=sci_arttext
reading_scope	bibliography-verified
problem_addressed	Interval-coded late-payment prediction for AR prioritization

data_method	ML classifier over invoice features into lateness intervals
limitations	South African industrial invoices; no censoring treatment
overlap_with_candidates	Supports aging-stage baseline design
proposed_difference	Censoring-aware competing-event labeling and refusal mechanism
verifiable_location	South African Journal of Industrial Engineering 33(4)

Row 9

source_id	PW09
title	Waste in the US Health Care System: Estimated Costs and Potential for Savings
authors	Shrank W.H.; Rogstad T.L.; Parekh N.
year	2019
url_doi	https://doi.org/10.1001/jama.2019.13978
reading_scope	bibliography-verified
problem_addressed	Administrative complexity as dominant waste source
data_method	Cost estimation study
limitations	Not a methods paper
overlap_with_candidates	Motivation for revenue-cycle analytics
proposed_difference	Not applicable
verifiable_location	JAMA 322(15):1501-1509